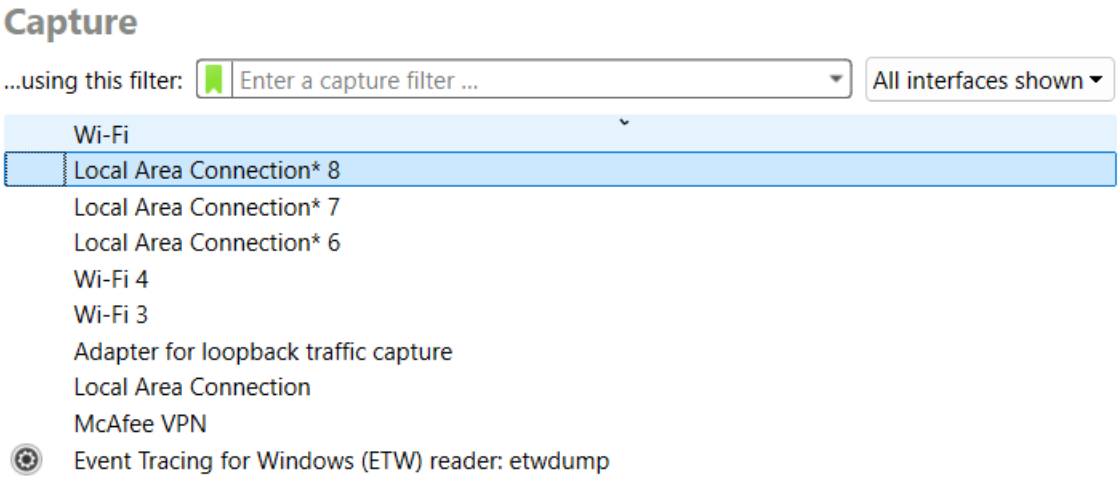
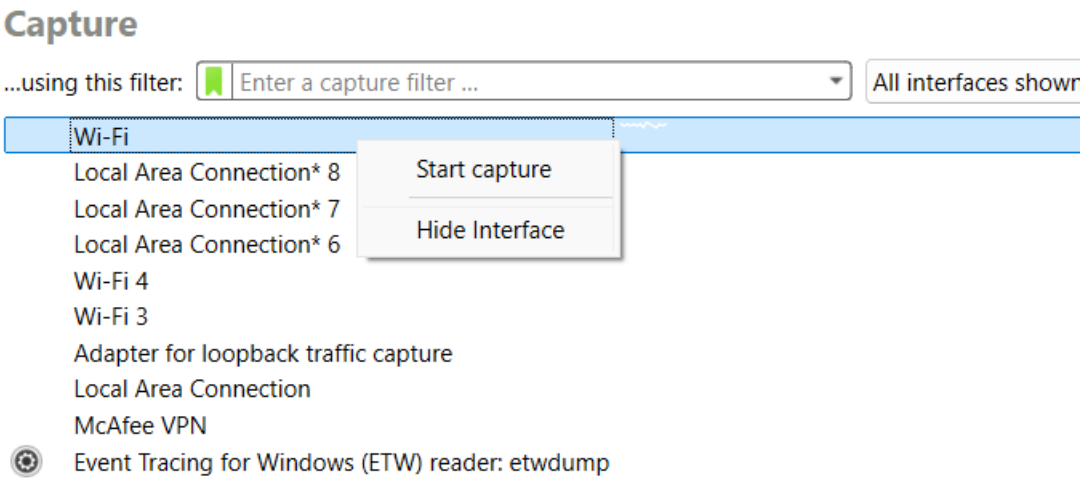
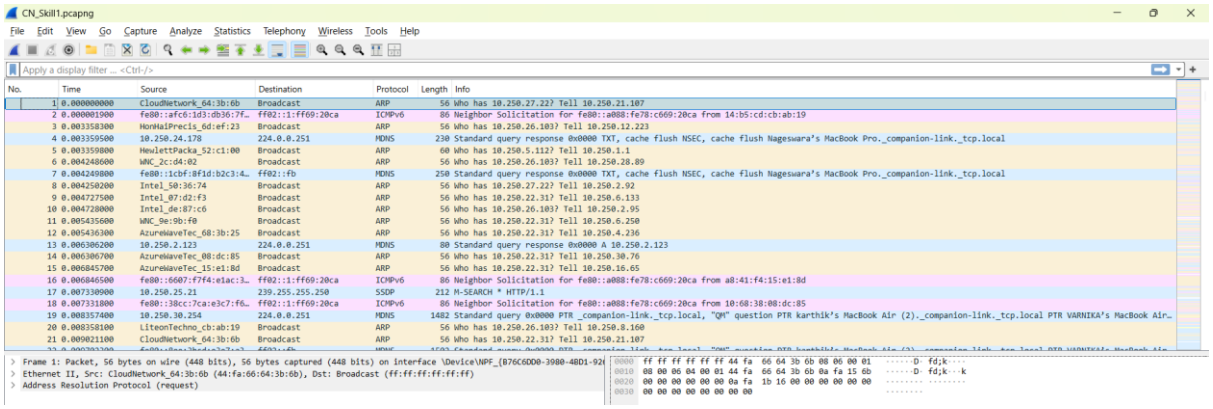
Skill – 1

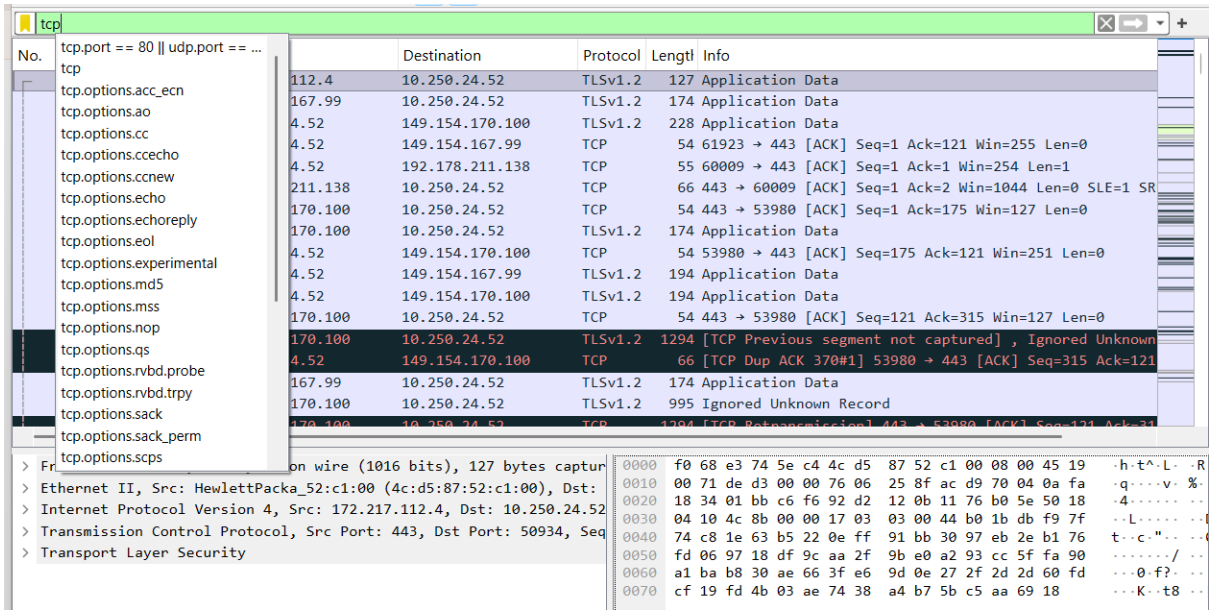
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

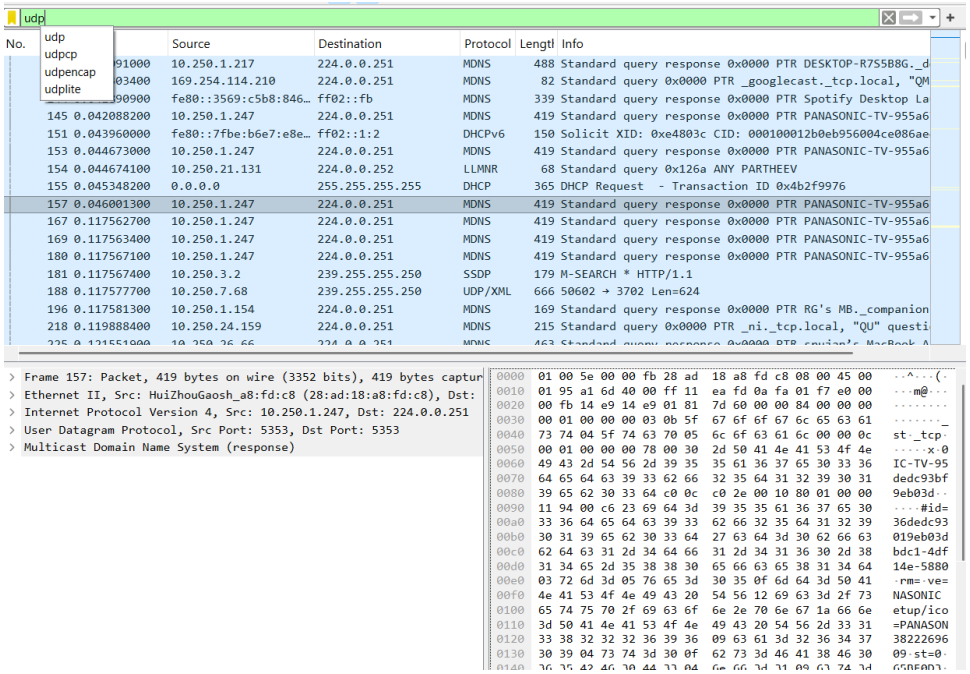
Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data

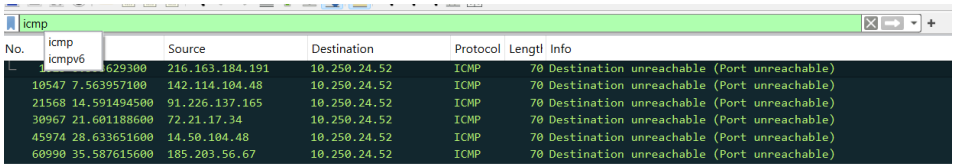


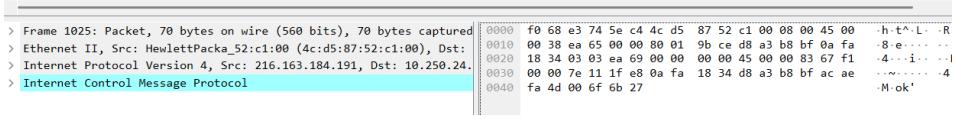


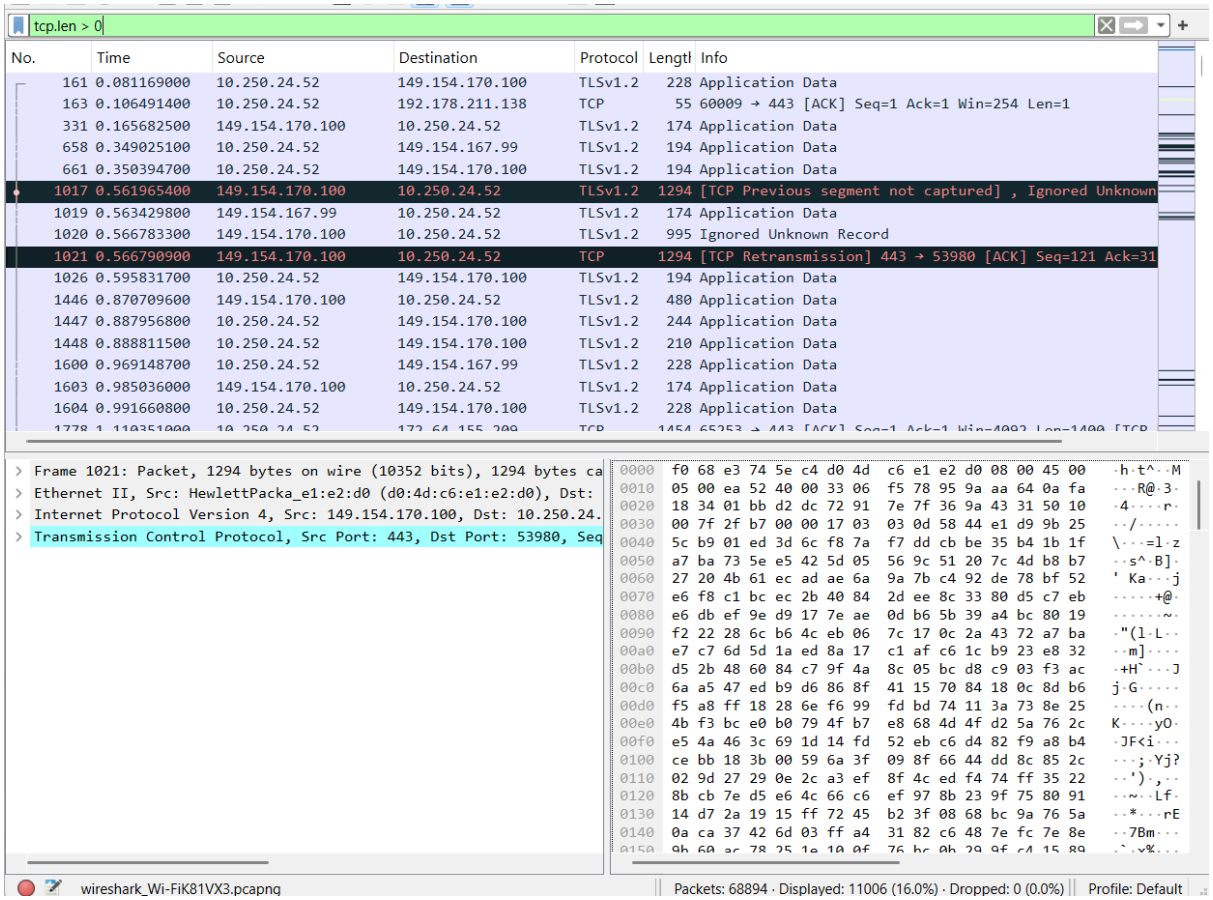

CN_Skill1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 CloudNetwork_64:3b:6b Broadcast ARP 56 Who has 10.250.27.22? Tell 10.250.21.107
2 0.000001900 fe80::afc6:1d3:db36:7f… ff02::1:ff69:20ca ICMPv6 86 Neighbor Solicitation for fe80::a088:fe78:c669:20ca from 14:b5:cd:cb:ab:19
3 0.003358300 HonHaiPrec1s_6d:ef:23 Broadcast ARP 56 Who has 10.250.26.103? Tell 10.250.12.223
4 0.003359500 10.250.24.178 224.0.0.251 MDNS 230 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Nageswara's MacBook Pro._companion-link._tcp.local
5 0.003359800 HewlettPacka_52:c1:00 Broadcast ARP 60 Who has 10.250.5.112? Tell 10.250.1.1
6 0.004248600 WNC_2c:d4:02 Broadcast ARP 56 Who has 10.250.26.103? Tell 10.250.28.89
7 0.004249800 fe80::1cbf:8f1d:b2c3:4… ff02::fb MDNS 250 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Nageswara's MacBook Pro._companion-link._tcp.local
8 0.004250200 Intel_50:36:74 Broadcast ARP 56 Who has 10.250.27.22? Tell 10.250.2.92
9 0.004727500 Intel_07:d2:f3 Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.6.133
10 0.004728000 Intel_de:87:c6 Broadcast ARP 56 Who has 10.250.26.103? Tell 10.250.2.95
11 0.005435600 WNC_9e:9b:f0 Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.6.250
12 0.005436300 AzureWaveTec_68:3b:25 Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.4.236
13 0.006306200 10.250.2.123 224.0.0.251 MDNS 80 Standard query response 0x0000 A 10.250.2.123
14 0.006306700 AzureWaveTec_08:dc:85 Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.30.76
15 0.006845700 AzureWaveTec_15:e1:8d Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.16.65
16 0.006846500 fe80::6607:f7f4:e1ac:3… ff02::1:ff69:20ca ICMPv6 86 Neighbor Solicitation for fe80::a088:fe78:c669:20ca from a8:41:f4:15:e1:8d
17 0.007330900 10.250.25.21 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
18 0.007331800 fe80::38cc:7ca:e3c7:f6… ff02::1:ff69:20ca ICMPv6 86 Neighbor Solicitation for fe80::a088:fe78:c669:20ca from 10:68:38:08:dc:85
19 0.008357400 10.250.30.254 224.0.0.251 MDNS 1482 Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR karthik's MacBook Air (2)._companion-link._tcp.local PTR VARNIKA's MacBook Air…
20 0.008358100 LiteonTechno_cb:ab:19 Broadcast ARP 56 Who has 10.250.26.103? Tell 10.250.8.160
21 0.009021100 CloudNetwork_64:3b:6b Broadcast ARP 56 Who has 10.250.22.31? Tell 10.250.21.107
Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF_{B76C6DD0-3980-4BD1-92(
Ethernet II, Src: CloudNetwork_64:3b:6b (44:fa:66:64:3b:6b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff 44 fa 66 64 3b 6b 08 06 00 01 ·····D· fd;k····
0010 08 00 06 04 00 01 44 fa 66 64 3b 6b 0a fa 15 6b ·····D· fd;k···k
0020 00 00 00 00 00 00 0a fa 1b 16 00 00 00 00 00 00 ···· ····
0030 00 00 00 00 00 00 00 00 ····


tcp
tcp.port == 80 || udp.port == ...
tcp
tcp.options.acc_ecn
tcp.options.ao
tcp.options.cc
tcp.options.ccecho
tcp.options.ccnew
tcp.options.echo
tcp.options.echoreply
tcp.options.eol
tcp.options.experimental
tcp.options.md5
tcp.options.mss
tcp.options.nop
tcp.options.qs
tcp.options.rvbd.probe
tcp.options.rvbd.trpy
tcp.options.sack
tcp.options.sack_perm
tcp.options.scps
No. Destination Protocol Lengtl Info
112.4 10.250.24.52 TLSv1.2 127 Application Data
167.99 10.250.24.52 TLSv1.2 174 Application Data
4.52 149.154.170.100 TLSv1.2 228 Application Data
4.52 149.154.167.99 TCP 54 61923 → 443 [ACK] Seq=1 Ack=121 Win=255 Len=0
4.52 192.178.211.138 TCP 55 60009 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
211.138 10.250.24.52 TCP 66 443 → 60009 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SR
170.100 10.250.24.52 TCP 54 443 → 53980 [ACK] Seq=1 Ack=175 Win=127 Len=0
170.100 10.250.24.52 TLSv1.2 174 Application Data
4.52 149.154.170.100 TCP 54 53980 → 443 [ACK] Seq=175 Ack=121 Win=251 Len=0
4.52 149.154.167.99 TLSv1.2 194 Application Data
4.52 149.154.170.100 TLSv1.2 194 Application Data
170.100 10.250.24.52 TCP 54 443 → 53980 [ACK] Seq=121 Ack=315 Win=127 Len=0
170.100 10.250.24.52 TLSv1.2 1294 [TCP Previous segment not captured] , Ignored Unknown
4.52 149.154.170.100 TCP 66 [TCP Dup ACK 370#1] 53980 → 443 [ACK] Seq=315 Ack=121
167.99 10.250.24.52 TLSv1.2 174 Application Data
170.100 10.250.24.52 TLSv1.2 995 Ignored Unknown Record
Fr on wire (1016 bits), 127 bytes captur
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst:
Internet Protocol Version 4, Src: 172.217.112.4, Dst: 10.250.24.52
Transmission Control Protocol, Src Port: 443, Dst Port: 50934, Seq
Transport Layer Security
0000 f0 68 e3 74 5e c4 4c d5 87 52 c1 00 08 00 45 19 ·h·t^·L· ·R····E·
0010 00 71 de d3 00 00 76 06 25 8f ac d9 70 04 0a fa ·q····v· %···p···
0020 18 34 01 bb c6 f6 92 d2 12 0b 11 76 b0 5e 50 18 ·4····· ···v·^P·
0030 04 10 4c 8b 00 00 17 03 03 00 44 b0 1b db f9 7f ··L····· ··D·····
0040 74 c8 1e 63 b5 22 0e ff 91 bb 30 97 eb 2e b1 76 t··c·"·· ··0····v
0050 fd 06 97 18 df 9c aa 2f 9b e0 a2 93 cc 5f fa 90 ·······/ ·····_··
0060 a1 ba b8 30 ae 66 3f e6 9d 0e 27 2f 2d 2d 60 fd ···0·f?· ··'/--`·
0070 cf 19 fd 4b 03 ae 74 38 a4 b7 5b c5 aa 69 18 ···K··t8 ··[··i·

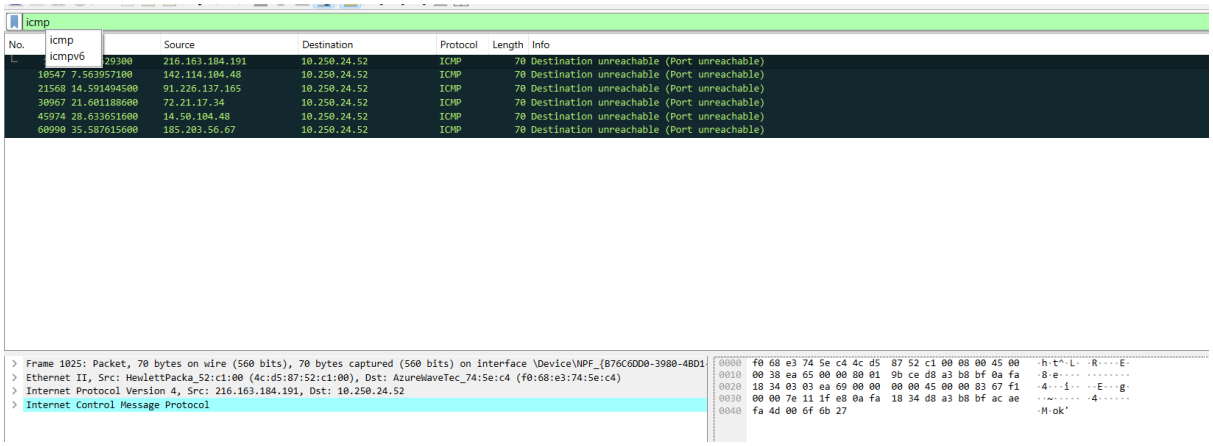

udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Lengtl Info
91000 10.250.1.217 224.0.0.251 MDNS 488 Standard query response 0x0000 PTR DESKTOP-R7S5B8G._d
03400 169.254.114.210 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM
90900 fe80::3569:c5b8:846… ff02::fb MDNS 339 Standard query response 0x0000 PTR Spotify Desktop La
145 0.042088200 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
151 0.043960000 fe80::7fbe:b6e7:e8e… ff02::1:2 DHCPv6 150 Solicit XID: 0xe4803c CID: 000100012b0eb956004ce086ae
153 0.044673000 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
154 0.044674100 10.250.21.131 224.0.0.252 LLMNR 68 Standard query 0x126a ANY PARTHEEV
155 0.045348200 0.0.0.0 255.255.255.255 DHCP 365 DHCP Request - Transaction ID 0x4b2f9976
157 0.046001300 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
167 0.117562700 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
169 0.117563400 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
180 0.117567100 10.250.1.247 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a6
181 0.117567400 10.250.3.2 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
188 0.117577700 10.250.7.68 239.255.255.250 UDP/XML 666 50602 → 3702 Len=624
196 0.117581300 10.250.1.154 224.0.0.251 MDNS 169 Standard query response 0x0000 PTR RG's MB._companion
218 0.119888400 10.250.24.159 224.0.0.251 MDNS 215 Standard query 0x0000 PTR _ni._tcp.local, "QU" questi
Frame 157: Packet, 419 bytes on wire (3352 bits), 419 bytes captur
Ethernet II, Src: HuiZhouGaosh_a8:fd:c8 (28:ad:18:a8:fd:c8), Dst:
Internet Protocol Version 4, Src: 10.250.1.247, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
0000 01 00 5e 00 00 fb 28 ad 18 a8 fd c8 08 00 45 00 ··^···(· ·····E·
0010 01 95 a1 6d 40 00 ff 11 ea fd 0a fa 01 f7 e0 00 ···m@··· ········
0020 00 fb 14 e9 14 e9 01 81 7d 60 00 00 84 00 00 00 ········ }`······
0030 00 01 00 00 00 03 0b 5f 67 6f 6f 67 6c 65 63 61 ·······_ googleca
0040 73 74 04 5f 74 63 70 05 6c 6f 63 61 6c 00 00 0c st·_tcp· local···
0050 00 01 00 00 00 78 00 30 2d 50 41 4e 41 53 4f 4e ·····x·0 -PANASON
0060 49 43 2d 54 56 2d 39 35 35 61 36 37 65 30 33 36 IC-TV-95 5a67e036
0070 64 65 64 63 39 33 62 66 32 35 64 31 32 39 30 31 dedc93bf 25d12901
0080 39 65 62 30 33 64 c0 0c c0 2e 00 10 80 01 00 00 9eb03d·· ········
0090 11 94 00 c6 23 69 64 3d 39 35 35 61 36 37 65 30 ····#id= 955a67e0
00a0 33 36 64 65 64 63 39 33 62 66 32 35 64 31 32 39 36dedc93 bf25d129
00b0 30 31 39 65 62 30 33 64 27 63 64 3d 30 62 66 63 019eb03d 'cd=0bfc
00c0 62 64 63 31 2d 34 64 66 31 2d 34 31 36 30 2d 38 bdc1-4df 1-4160-8
00d0 31 34 65 2d 35 38 38 30 65 66 63 65 38 31 34 64 14e-5880 efce814d
00e0 03 72 6d 3d 05 76 65 3d 30 35 0f 6d 64 3d 50 41 ·rm=·ve= 05·md=PA
00f0 4e 41 53 4f 4e 49 43 20 54 56 12 69 63 3d 2f 73 NASONIC TV·ic=/s
0100 65 74 75 70 2f 69 63 6f 6e 2e 70 6e 67 1a 66 6e etup/ico n.png·fn
0110 3d 50 41 4e 41 53 4f 4e 49 43 20 54 56 2d 33 31 =PANASON IC TV-31
0120 33 38 32 32 32 36 39 36 09 63 61 3d 32 36 34 37 38222696 ·ca=2647
0130 30 39 04 73 74 3d 30 0f 62 73 3d 46 41 38 46 30 09·st=0· bs=FA8F0
0140 3C 35 42 4C 30 44 33 04 6e 6e 3d 31 09 63 74 3d <5BL0D3· nn=1·ct=
GSDE0D1·


icmp
icmp
icmpv6
No. Source Destination Protocol Lengtl Info
629300 216.163.184.191 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
10547 7.563957100 142.114.104.48 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
21568 14.591494500 91.226.137.165 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
30967 21.601188600 72.21.17.34 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
45974 28.633651600 14.50.104.48 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
60990 35.587615600 185.203.56.67 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)


Frame 1025: Packet, 70 bytes on wire (560 bits), 70 bytes captured
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst:
Internet Protocol Version 4, Src: 216.163.184.191, Dst: 10.250.24.
Internet Control Message Protocol
0000 f0 68 e3 74 5e c4 4c d5 87 52 c1 00 08 00 45 00 ·h·t^·L· ·R····E·
0010 00 38 ea 65 00 00 80 01 9b ce d8 a3 b8 bf 0a fa ·8·e···· ····
0020 18 34 03 03 ea 69 00 00 00 00 45 00 00 83 67 f1 ·4···i·· ··E···g·
0030 00 00 7e 11 1f e8 0a fa 18 34 d8 a3 b8 bf ac ae ··~····· ·4····
0040 fa 4d 00 6f 6b 27 ·M·ok'



Parameter	TCP Packet	UDP Packet
Frame Length	1294 bytes	365 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	1280 bytes	351 bytes
Payload Length	1240 bytes	323 bytes


icmp
icmp
icmpv6
No.
Source
Destination
Protocol
Length
Info
29300 216.163.184.191 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
10547 7.563957100 142.114.104.48 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
21568 14.591494500 91.226.137.165 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
30967 21.601188600 72.21.17.34 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
45974 28.633651600 14.50.104.48 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
60990 35.587615600 185.203.56.67 10.250.24.52 ICMP 70 Destination unreachable (Port unreachable)
Frame 1025: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface \Device\NPF_{B76C6DD0-3980-4BD1
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: AzureWaveTec_74:5e:c4 (f0:68:e3:74:5e:c4)
Internet Protocol Version 4, Src: 216.163.184.191, Dst: 10.250.24.52
Internet Control Message Protocol
0000 f0 68 e3 74 5e c4 4c d5 87 52 c1 00 08 00 45 00
0010 00 38 ea 65 00 00 80 01 9b ce d8 a3 b8 bf 0a fa
0020 18 34 03 03 ea 69 00 00 00 00 45 00 00 83 67 f1
0030 00 00 7e 11 1f e8 0a fa 18 34 d8 a3 b8 bf ac ae
0040 fa 4d 00 6f 6b 27